

PROJECT SYNOPSIS

ON

MEV Aware DeFi Liquidation System

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Engineering-AIML**

**in Partial Fulfillment of the Requirements
for the Degree of**

Bachelor of Technology

in

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UTTAR PRADESH, LUCKNOW

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C410.4: Work effectively as a part of team in multi-disciplinary areas.

C410.5: Consolidate the final outcome in the form of a publication.

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	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C410.1	3	3	3	3	2	3	3	3	3	3	2	3	3	3
C410.2	2	2	2	2	2	2	2	2	2	3	2	3	2	3
C410.3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
C410.4	3	3	3	3	2	3	2	3	3	3	3	3	3	3
C410.5	3	3	3	3	3	3	3	3	3	3	3	3	3	3
C410	2.80	2.80	2.80	2.80	2.40	2.80	2.40	2.80	2.80	3.00	2.60	3.00	2.80	3.00

DECLARATION

We hereby declare that this synopsis submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that project synopsis report entitled “.....”
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MEV Aware DeFi Liquidation System

ABSTRACT

Decentralized finance (DeFi) lending protocols rely on automated liquidations to maintain protocol solvency when collateral values fall below defined health thresholds. In public blockchain networks like Ethereum, liquidation calls submitted to the public mempool expose transactions to predatory Maximal Extractable Value (MEV) extraction, such as front-running, sandwich attacks, and priority gas auctions (PGAs). These competitive auctions consume disproportionate network bandwidth, drive up transaction fees, and induce latency that often leads to liquidation failure and bad debt accumulation for protocols.

This project proposes an MEV-Aware DeFi Liquidation System designed to execute undercollateralized debt liquidations efficiently while mitigating negative externalities associated with public mempool competition. The proposed framework integrates private transaction order flows, dynamic auction routing, and predictive health-factor monitoring powered by machine learning algorithms. By analyzing historical mempool dynamics, on-chain volatility, and slippage thresholds, the system schedules and bundles liquidations to bypass public searcher races. The expected outcome is a robust, low-latency liquidation mechanism that minimizes value leakage to searchers, preserves collateral recovery value for borrowers, and enhances protocol solvency across high-volatility market regimes.

Keywords: Decentralized Finance (DeFi), Maximal Extractable Value (MEV), Lending Protocols, Liquidations, Front-running, Solvency, Machine Learning.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Acronym	Description
MEV	Maximal Extractable Value
DeFi	Decentralized Finance
DEX	Decentralized Exchange
AMM	Automated Market Maker
PGA	Priority Gas Auction
LTV	Loan-to-Value Ratio
HF	Health Factor
RPC	Remote Procedure Call
PnL	Profit and Loss
EVM	Ethereum Virtual Machine
PBS	Proposer-Builder Separation

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CHAPTER 1

INTRODUCTION

1.1 Background and DeFi Lending

Decentralized Finance (DeFi) lending platforms such as Aave, Compound, and MakerDAO allow users to deposit cryptocurrency assets as collateral and borrow other digital assets against them. Unlike traditional banking systems that evaluate borrower creditworthiness through centralized credit scores or legal recourse, DeFi lending protocols operate entirely on overcollateralization. Borrowers must deposit collateral whose market value substantially exceeds the principal amount borrowed.

To continuously gauge loan solvency, protocols calculate a composite metric termed the **Health Factor** (HF):

$$HF = \frac{\sum_i (C_i \times P_i \times LT_i)}{\sum_j (D_j \times P_j)} \quad (1.1)$$

where C_i represents the quantity of deposited collateral asset i , P_i denotes its current oracle price, LT_i denotes the asset-specific liquidation threshold (typically 75%–85%), and $D_j \times P_j$ represents the total outstanding debt evaluated in reference currency.

As long as $HF > 1.0$, the position remains solvent. However, when asset price volatility causes the collateral value to fall or the borrowed asset to appreciate, the health factor drops below unity ($HF < 1.0$). At this exact threshold, the loan is designated as undercollateralized, and the protocol permits external participants—known as liquidators—to repay a fraction of the outstanding debt in exchange for seizing borrower collateral at a discounted rate.

1.2 The Problem with Public Mempool Liquidations

In conventional protocols, liquidation operates on a first-come, first-served execution model. The first valid transaction included in a block successfully liquidates the debt and claims the liquidation bonus (typically 5% to 10%). Because this bonus can translate into thousands of dollars in risk-free profit, dozens of automated searcher bots compete simultaneously for the exact same liquidation opportunity.

In public blockchain networks such as Ethereum, pending transactions reside in a transparent mempool before inclusion. When searchers detect an undercollateralized position, they engage in **Priority Gas Auctions (PGAs)**. Each bot observes competitor bids and repeatedly broadcasts replacement transactions with higher priority gas fees to incentivize block validators to place their transaction first.

This competitive dynamic triggers severe systemic externalities:

- **Economic Value Leakage:** Competitive searchers surrender 70% to 90% of their gross liquidation profit directly to block builders and validators as gas bribes, diverting capital away from the lending ecosystem.
- **Unnecessary Borrower Collateral Loss:** Fixed liquidation discounts penalize borrowers heavily, regardless of whether secondary market conditions actually require such discounts to achieve liquidation.
- **Mempool Congestion and Failed Transactions:** Thousands of competing liquidation transactions fail every day, consuming block space and driving up baseline gas costs for ordinary network users.
- **Cascading Market Slippage:** Liquidators typically dump seized collateral immediately on Automated Market Makers (AMMs) like Uniswap to lock in profits, depressing market prices and triggering secondary liquidations across neighboring protocols.

1.3 Proposed Solution: The Reforge Architecture

This project develops and evaluates **Reforge**, an MEV-aware batch auction liquidation mechanism for DeFi lending protocols. Instead of running an uncoordinated race in the public

mempool, Reforge aggregates competing liquidator bids within a discrete, short time window (1–2 blocks) and evaluates them based on **Net Economic Value** rather than the size of a gas bribe.

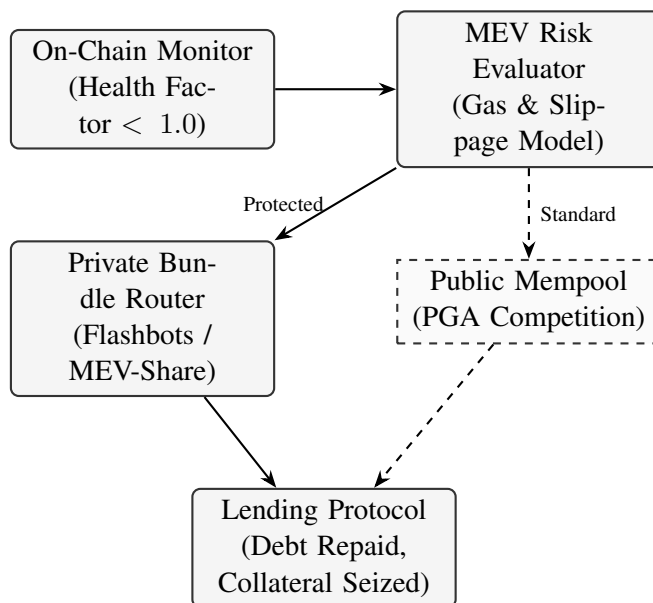


Figure 1.1: Transaction flow comparison between private MEV-aware bundling and standard public mempool liquidations.

Figure 1.1 compares standard public mempool execution against our MEV-aware pipeline. By establishing an orderly auction process, Reforge aligns incentives across all three primary stakeholders: borrowers retain more residual equity, protocols prevent bad debt accumulation through deterministic single-transaction settlement, and liquidators earn predictable margins based on execution quality rather than gas expenditure.

CHAPTER 2

MOTIVATION

2.1 The Failure of Priority Gas Auctions

The primary motivation behind this research stems from the structural inefficiencies of first-come, first-served liquidations in transparent blockchain networks. When a borrower's account falls below $HF = 1.0$, the protocol creates an open-access profit opportunity. Multiple independent searcher bots identify the position within the same second and race to submit liquidation transactions.

Because blockchain miners and Proof-of-Stake block proposers order transactions within a block by descending priority fee (gas price), searchers enter a Priority Gas Auction (PGA). A typical sequence unfolds as follows:

1. **Bot A** detects the liquidation with an expected gross profit of \$500 and submits a transaction offering a 20 Gwei priority fee.
2. **Bot B** detects the same opportunity and broadcasts a replacement transaction offering 45 Gwei.
3. **Bot C** outbids with 90 Gwei, prompting Bot B to make a final push at 140 Gwei.
4. The block builder includes Bot B's transaction at the top of the block and rejects or fails all competing bids.

Figure 2.1 illustrates this competitive escalation and value capture in the public mempool.

As summarized in Table 2.1, the winning liquidator surrendered \$220 (44% of gross margin) to the block builder, while the losing bots burned gas on reverted calls. The borrower bore the full economic cost of this \$500 penalty, but the majority of the surplus was extracted by consensus intermediaries rather than utilized to secure the lending platform.

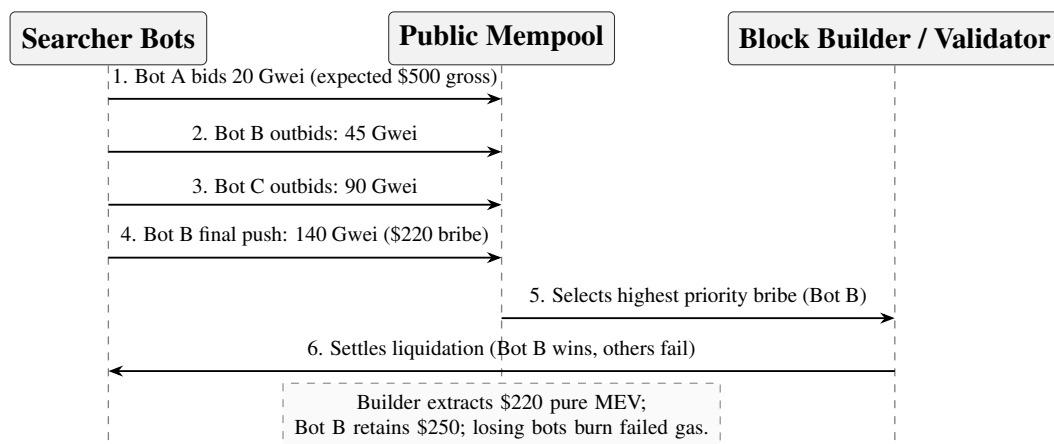


Figure 2.1: Sequence of a Priority Gas Auction showing competitive gas escalation and validator value capture.

Table 2.1: Economic breakdown of a typical Priority Gas Auction liquidation.

Participant	Gross Profit	Gas Paid	DEX Slippage	Net Profit
Bot A (Failed)	\$500.00	\$40.00	\$0.00	-\$40.00
Bot B (Winner)	\$500.00	\$220.00	\$30.00	\$250.00
Bot C (Failed)	\$500.00	\$150.00	\$0.00	-\$150.00
Validator / Builder	—	+\$220.00	—	+\$220.00

2.2 Excessive Borrower Collateral Loss

Traditional lending protocols impose a fixed liquidation penalty, typically 5% to 10% of the repaid debt. This fixed discount is arbitrary: whether the market for the underlying collateral is highly liquid with tight spreads, or illiquid during an extreme crash, the distressed borrower is penalized by the exact same percentage.

Empirical studies by Qin et al. (2021) and Tian and Xu (2025) demonstrate that fixed-spread protocols force borrowers to forfeit substantial equity that could have otherwise been preserved. In contrast, competitive auction mechanisms allow liquidators to bid for the minimum required discount necessary to clear debt. Data from the Bank of Canada (2025) shows that transitioning from fixed discounts to competitive auctions reduces average liquidation discounts from 5.0% to 1.8%—a 64% reduction in unnecessary borrower loss.

2.3 Secondary Market Slippage and Cascades

Liquidators rarely hold seized collateral long-term; they instantly sell it on decentralized exchanges (such as Uniswap or Curve) to repay the flash loans utilized to finance the liquidation. During market downturns, dozens of liquidation sales hit the same Automated Market Maker (AMM) pools within a short window. This causes localized price slippage, driving the oracle price lower and triggering secondary waves of liquidations across neighboring platforms.

2.4 Builder Monopolies and Access Centralization

With the advent of Proposer-Builder Separation (PBS) through MEV-Boost, MEV extraction has shifted from public mempools into private builder markets. Recent empirical measurements by Wang et al. (2024) show that the top three block builders construct over 95% of all Ethereum blocks. Furthermore, vertically integrated searcher-builders enjoy structural latency and informational advantages over independent liquidators. This centralization undermines the decentralized ethos of DeFi and creates single points of economic failure.

2.5 Core Research Questions

To address these compounding challenges, this research investigates two central questions:

1. **Research Question 1:** Does competitive MEV extraction in public mempools make DeFi liquidations unnecessarily expensive and amplify borrower losses?
2. **Research Question 2:** Can the liquidation mechanism be redesigned so that competition is evaluated on net economic efficiency rather than a race to submit the highest gas bribe?

CHAPTER 3

OBJECTIVE(S)

3.1 Primary Objective

The primary objective of this project is to design, implement, and empirically validate an end-to-end MEV-aware batch auction mechanism for decentralized lending protocols. The proposed system, named **Reforge**, replaces latency-sensitive Priority Gas Auctions with an objective economic efficiency metric, selecting winning liquidators by **Net Economic Value** rather than the magnitude of their gas bribes.

3.2 Specific Technical and Research Goals

To realize the primary objective, the project pursues five specific technical milestones:

1. **Mathematical Formulation of the Scoring Metric:** Develop an objective scoring function that quantifies the net economic utility generated by competing liquidation bids:

$$NEV = V_{\text{gross}} - C_{\text{gas}} - C_{\text{slippage}} \quad (3.1)$$

where V_{gross} represents the nominal liquidation bonus value, C_{gas} denotes the actual computational execution cost, and C_{slippage} accounts for secondary decentralized exchange price impact. Figure 3.1 illustrates the end-to-end auction workflow.

2. **Smart Contract Architecture Implementation:** Design and deploy modular Solidity smart contracts on an EVM-compatible execution environment using Foundry:
 - `LendingPool.sol`: Manages collateral deposits, borrowing accounting, and interest rate accrual.

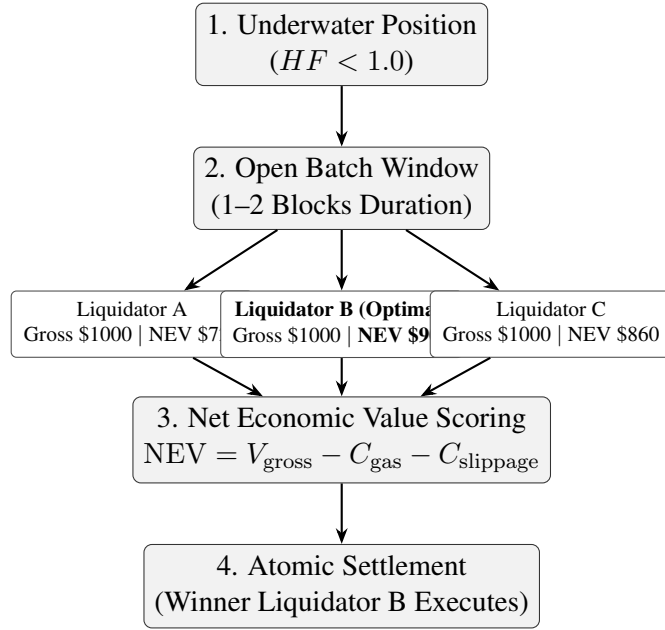


Figure 3.1: Reforge batch auction workflow from unhealthy position detection to optimal single-transaction settlement.

- `LiquidationAuction.sol`: Coordinates discrete 1–2 block auction windows, registers liquidator commitments, and executes atomic settlements.
 - `HealthFactorLib.sol`: Provides optimized, gas-efficient on-chain health factor calculations.
3. **Real-Time Asynchronous Backend Infrastructure:** Construct a low-latency backend coordinator in Node.js and TypeScript utilizing Fastify, viem, and Redis/BullMQ to subscribe to blockchain block headers, identify unhealthy loan positions ($HF < 1.0$), ingest liquidator bids, and dispatch winning bundles.
 4. **Web Portal and Liquidator Marketplace:** Build a responsive web application using Next.js 16, React 19, and Tailwind CSS to display protocol solvency metrics, real-time liquidation auctions, and an interactive simulation laboratory for researchers.
 5. **Empirical Simulation and Comparative Evaluation:** Construct a Python-based simulation engine to run Monte Carlo market stress tests, comparing Reforge against baseline PGA protocols across key performance indicators: borrower collateral retention, liquidator profit margins, total network gas expenditure, and auction settlement latency.

3.3 Tripartite Stakeholder Alignment

A foundational goal of Reforge is achieving balanced incentives across all three primary DeFi participants, as summarized in Table 3.1.

Table 3.1: Incentive alignment comparison between conventional systems and Reforge.

Stakeholder	Conventional System (PGA)	Reforge Batch Auction
Borrower	Forfeits static 5%–10% collateral penalty regardless of market depth.	Retains residual equity by clearing debt at dynamic, competitive discounts.
Protocol	High transaction failure rates during mempool spikes lead to bad debt.	Guaranteed single-transaction execution ensures reliable debt recovery.
Liquidator	Surrenders 70%–90% of gross profit to block builders in gas wars.	Captures sustainable profit margins based on routing and DEX efficiency.

CHAPTER 4

SCOPE OF THE PROJECT

4.1 System Architecture and Layer Decomposition

The scope of the Reforge project spans four interconnected technological tiers, ensuring a comprehensive implementation from core smart contracts to empirical research tooling:

1. **Smart Contract Execution Layer:** Implements decentralized financial logic in Solidity ($\geq 0.8.24$) using the Foundry toolchain. The layer manages user balances, computes collateral health factors, enforces the discrete auction lifecycle, and executes flash repayments and collateral payouts atomically.
2. **Backend Coordination and Event Pipeline:** Developed in Node.js and TypeScript using the Fastify framework. It maintains persistent WebSocket connections to the blockchain RPC, detects blocks where positions breach $HF = 1.0$, manages auction state transitions using Redis and BullMQ, and evaluates competing liquidator bids using the Net Economic Value scoring function.
3. **Frontend User Interface and Telemetry Portal:** Built with Next.js 16, React 19, and Tailwind CSS. The interface provides end-users with collateral health dashboards, presents liquidators with a real-time auction marketplace, and offers researchers interactive visualization tools.
4. **Empirical Research and Simulation Engine:** Constructed in Python 3.12 utilizing NumPy, Pandas, and Matplotlib. It simulates synthetic price shocks, replays historical Ethereum market downturns, and benchmarks the economic outcomes of batch auctions against Priority Gas Auctions.

Figure 4.1 illustrates the cross-tier dataflow and interaction among these four operational components.

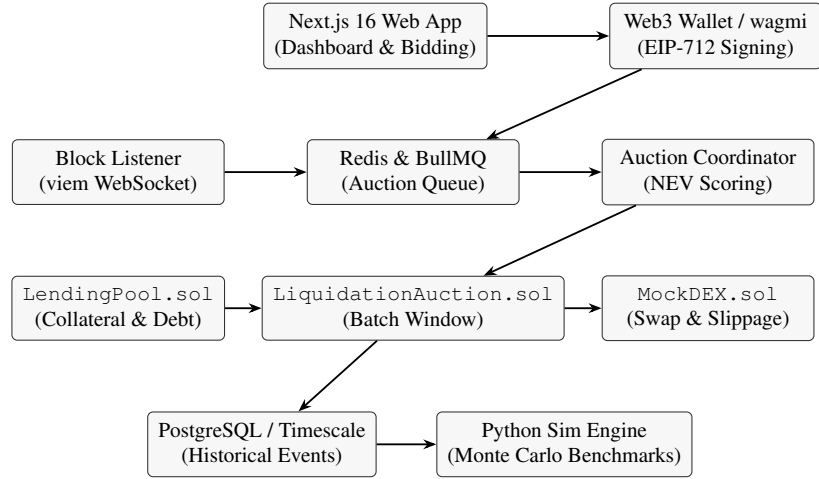


Figure 4.1: End-to-end multi-tier system architecture of the Reforge liquidation platform.

4.2 Operational Assumptions and Boundaries

To maintain a well-defined project scope, Reforge operates under specific foundational parameters:

- **Target Blockchain Environment:** The protocol is designed for Ethereum Mainnet and EVM-compatible Layer-2 rollups (such as Arbitrum and Optimism) that maintain deterministic block execution.
- **Oracle Architecture:** Asset valuations rely on external decentralized price oracles (such as Chainlink) with standard heartbeat and deviation update thresholds.
- **Auction Window Duration:** The discrete batch auction window is bounded between 1 and 2 blocks (approximately 12 to 24 seconds on Ethereum L1). This duration balances adequate liquidator participation against the risk of rapid price depreciation.

4.3 The Latency vs. Solvency Trade-off

A central analytical challenge within the project scope is evaluating the **Auction Latency Problem**. In conventional protocols, liquidation is continuous and instantaneous: the first transaction submitted in the next block clears the loan. Reforge introduces an explicit time window of 1–2 blocks to collect and score competing bids.

During this brief window, if the collateral price experiences an extreme flash crash, the position could deteriorate into negative equity (where debt value exceeds total collateral

value) before settlement occurs. Quantifying this exact trade-off—whether the massive reduction in MEV gas waste and preserved borrower equity outweighs the marginal risk of auction latency—constitutes a core focus of our research simulation.

4.4 Out-of-Scope Boundaries

The following aspects are explicitly excluded from the current project scope:

- **Cross-Chain Atomic Settlement:** Multi-chain liquidations spanning distinct consensus layers via bridges are excluded; all settlement occurs natively on a single EVM state machine.
- **Under-Collateralized Lending:** The system focuses exclusively on overcollateralized lending pools; identity-based credit scoring or zero-collateral lending is out of scope.
- **Proprietary Off-Chain Bot Strategies:** The project specifies the protocol-level scoring mechanism and reference liquidator interfaces, but does not encompass proprietary high-frequency trading arbitrage algorithms.

CHAPTER 5

RELATED WORK

The design and evaluation of Reforge builds directly upon recent academic breakthroughs spanning DeFi liquidation mechanics, Maximal Extractable Value (MEV), Proposer-Builder Separation (PBS), and consensus-level order-fairness.

5.1 Empirical Studies on DeFi Liquidations

A critical pillar of lending protocol stability is understanding how liquidation mechanisms behave under real market stress:

- **Qin et al. (IMC 2021):** Conducted the first large-scale empirical evaluation of liquidations across Aave, Compound, MakerDAO, and dYdX [2]. The authors demonstrated that static liquidation discounts (5%–10%) transfer oversized economic surplus from distressed borrowers to liquidators. Even in highly liquid market environments with minimal execution frictions, borrowers were forced to forfeit the full fixed penalty. Furthermore, they proved that liquidators strategically optimize partial repayments and gas bids to maximize extraction.
- **Tian and Xu (Bank of Canada 2025):** Provided direct empirical comparison between fixed-spread models (Aave) and competitive Dutch auctions (MakerDAO) [4]. Their findings reveal that competitive auctions reduce average liquidation discounts from 5.0% to 1.8% (a 64% reduction) and cut secondary market price impact by 81% (from 0.054% to 0.010%). Their analysis highlights two competing economic forces: the *participation effect* (auctions attract more diverse searchers) and the *competition effect* (bidding drives clearance prices closer to fair market value).

5.2 MEV Foundations and Priority Gas Auctions

The theoretical and empirical understanding of transaction reordering in transparent mempools is rooted in seminal MEV research:

- **Daian et al. (Flash Boys 2.0, IEEE S&P 2019):** First formalized Maximal Extractable Value and demonstrated that pending transactions in public mempools invite automated Priority Gas Auctions (PGAs) [1]. The authors proved that when extractable value is large relative to block subsidies, rational miners are economically incentivized to execute “time-bandit attacks”—reorganizing historical blocks to steal past MEV opportunities.
- **Qin et al. (Quantifying BEV, IEEE S&P 2022):** Measured over \$540.54 million in extracted MEV across 32 months of Ethereum mainnet data [3]. The study demonstrated that liquidations constitute 26% of all extracted value (second only to DEX arbitrage), with single liquidation events yielding profits up to \$4.1 million.

5.3 Proposer-Builder Separation and Private Flow

With the introduction of MEV-Boost following Ethereum’s transition to Proof-of-Stake, the structural dynamics of MEV shifted into specialized builder auctions:

- **Pai and Resnick (2023):** Analyzed the structural advantages enjoyed by vertically integrated searcher-builders in MEV-Boost auctions [5]. They proved that integrated builders observe competing bundles with zero external communication latency and effectively participate in second-price auction dynamics, while independent liquidators face the “winner’s curse”.
- **Wang et al. (2024):** Empirically measured private order flow dynamics on Ethereum, demonstrating that the top three block builders construct over 95% of all blocks [6]. This market concentration proves that private builder channels do not eliminate unfair competition, but rather centralize it.
- **Natale and Moser (2023):** Investigated artificial latency games in PBS, showing that

block proposers systematically delay block proposals by 500–1000ms to extract a median 1.28% increase in MEV bribes [7].

Figure 5.1 illustrates the modern MEV supply chain operating between searchers, block builders, and consensus validators under PBS.

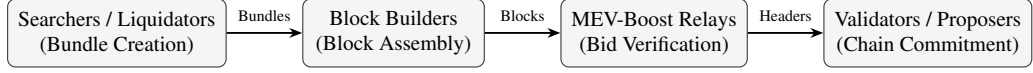


Figure 5.1: The modern MEV supply chain under Proposer-Builder Separation (PBS).

5.4 Order-Fairness and Security Taxonomies

- **Kelkar et al. (CRYPTO 2020):** Introduced the concept of *order-fairness* for Byzantine consensus protocols (the Aequitas family) [8], proving that standard consensus safety and liveness guarantees do not protect against sequence manipulation by block proposers.
- **Eskandari et al. (FC 2019):** Formulated the canonical front-running taxonomy: displacement, insertion, and suppression [9]. Standard mempool liquidation races represent pure displacement attacks, where searchers copy and outbid competitors.
- **Werner et al. (2021) and Gogol et al. (2024):** Systematized DeFi risk frameworks [10, 11], demonstrating that lending protocol insolvency arises from external economic incentive failures and mempool latency games rather than smart contract coding bugs.

5.5 Comparative Synthesis Matrix

Table 5.1 synthesizes how Reforge compares against prevailing liquidation architectures in decentralized finance.

5.6 Conclusion

The literature confirms that while conventional liquidation systems successfully maintain protocol solvency during benign market conditions, their reliance on first-come public mem-

Table 5.1: Architectural comparison of liquidation mechanisms across DeFi protocols.

Protocol	Mechanism Type	Pricing Model	MEV Resistance	Borrower Equity
Aave v3	First-come Mem-pool	Fixed Penalty (5–10%)	None (PGA Gas Wars)	Low (Static Loss)
Compound v3	First-come Mem-pool	Fixed Penalty (5–8%)	None (PGA Gas Wars)	Low (Static Loss)
MakerDAO	Dutch Auction	Descending Price Curve	Moderate (Latency Bidding)	Medium (Dynamic)
Reforge	Batch Auction	Net Economic Value	High (Order-Fair)	High (Maximized)

pool races imposes severe costs: excessive borrower collateral loss, catastrophic gas wars, and block-builder centralization. Existing auction designs, while superior to fixed spreads, remain susceptible to latency exploitation and builder monopolies. This establishes a clear need for Reforge’s application-level batch clearing mechanism.

5.7 Future Scope

Future extensions of this work include integrating zero-knowledge proofs (ZKPs) for private bid commitments, extending the batch clearing model to cross-chain lending protocols via LayerZero or Chainlink CCIP, and applying reinforcement learning algorithms to dynamically adjust the discrete auction window duration in response to real-time market volatility.

CHAPTER 6

HARDWARE & SOFTWARE REQUIREMENTS

The development, deployment, and simulation of the Reforge platform require a robust engineering stack spanning blockchain execution clients, high-throughput backend services, interactive user interfaces, and scientific research environments.

6.1 Software Requirements

The software ecosystem is decomposed into four functional tiers, as detailed in Table 6.1.

6.2 Hardware Requirements

Executing local EVM test forks, running asynchronous WebSocket listeners, and processing large-scale Monte Carlo price trajectory simulations necessitate high-performance workstation hardware. Table 6.2 outlines the minimum and recommended specifications.

Table 6.1: Comprehensive software technology stack for the Reforge protocol.

Subsystem	Technology	Version	Role & Functionality
Smart Contracts	Solidity	$\geq 0.8.24$	Core contract logic with transient storage support.
	Foundry (forge)	Latest	Compilation, fuzzing, and invariant testing.
	Anvil	Local Node	High-speed local EVM node with state manipulation.
	OpenZeppelin	v5.0	Standard ERC-20, SafeERC20, and ReentrancyGuard.
Backend	Node.js	v22 LTS	Asynchronous execution runtime for I/O event loops.
	TypeScript	v5.x	Strict static typing across API and RPC schemas.
	Fastify	v4.x	Low-overhead REST API server for liquidator bids.
	viem	v2.x	Lightweight Ethereum client with WebSocket listeners.
	Redis & BullMQ PostgreSQL	v7.x v16	In-memory job queue for auction timers and events. Relational storage for historical bids and telemetry.
Frontend Views	Next.js	16.3.0	React 19 App Router with Turbopack compilation.
	Tailwind CSS wagmi / viem	v4.0 Latest	Utility-first responsive styling engine. React hooks for Web3 wallet connection and signing.
Research & Sim	Python	3.12	Data science, numerical modeling, and simulation.
	NumPy & SciPy	Latest	Vectorized health factor and price trajectory models.
	Pandas	Latest	Tabular data processing for multi-run auction logs.
	Matplotlib	Latest	Publication-quality charts for empirical evaluation.

Table 6.2: Hardware specifications for development and simulation nodes.

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Quad-core Intel Core i5 or Ryzen 5	8-core Intel Core i7/i9 or Ryzen 7/9
System Memory	16 GB DDR4	32 GB DDR5 (for in-memory simulation)
Storage (SSD)	256 GB NVMe SSD	1 TB NVMe PCIe 4.0 SSD
Network	50 Mbps broadband	100+ Mbps low-latency fiber
Operating System	Windows 11 / Ubuntu 22.04 LTS	Ubuntu 24.04 LTS or WSL2

CHAPTER 7

PREDICTED TIMELINE FOR THE PROJECT

The implementation and scientific evaluation of Reforge are structured across an eight-month academic timeline, partitioned into six distinct engineering and research phases.

7.1 Phase Breakdown and Sprint Structure

The project employs an iterative development methodology with bi-weekly development sprints to ensure continuous integration, testing, and empirical feedback:

- **Phase 1: Literature Review & Mathematical Modeling (Months 1–2):** Comprehensive synthesis of academic literature on MEV, Proposer-Builder Separation, and DeFi liquidation mechanisms. Formal derivation and parameter tuning of the Net Economic Value (NEV) scoring equation.
- **Phase 2: Smart Contract Core Development (Months 3–4):** Implementation of the core Solidity contracts (`LendingPool.sol`, `LiquidationAuction.sol`, `HealthFactorLib.sol`). Writing automated unit tests, fuzz testing suites, and invariant checks in Foundry.
- **Phase 3: Real-Time Backend Coordination Pipeline (Months 4–5):** Constructing the Node.js/TypeScript backend services, including WebSocket RPC block subscription, Redis/BullMQ opportunity scheduling, and the off-chain bid ingestion and scoring coordinator.
- **Phase 4: Frontend Web Portal & Marketplace Terminal (Months 5–6):** Developing the Next.js 16 user interface, integrating wallet connection hooks via wagmi/viem, rendering live liquidation auctions, and building the interactive research telemetry console.

- **Phase 5: Empirical Simulation & Benchmark Experiments (Months 6–7):** Constructing the Python-based Monte Carlo market simulation engine. Running stress tests under historical market shock conditions (e.g., March 2020 Black Thursday, May 2022 Terra-Luna collapse) to evaluate collateral retention and latency trade-offs against baseline PGA mechanisms.
- **Phase 6: Analysis, Documentation & Thesis Compilation (Months 7–8):** Synthesizing simulation results, drafting academic research papers, completing formal thesis documentation, and preparing the final project defense presentation.

7.2 Milestone Schedule

Table 7.1 details the projected milestone deliverables across each phase.

Table 7.1: Detailed project milestones and timeline schedule.

Phase	Milestone Task	Timeline	Key Deliverable
1	Literature Survey & NEV Formulation	Month 1–2	Literature review synthesis and formal scoring specification.
2	Smart Contract Implementation	Month 3–4	Foundry test suite with >95% code coverage and invariant tests.
3	Real-Time Backend Engine	Month 4–5	Functional WebSocket listener and Redis queue scoring service.
4	Next.js Web Interface	Month 5–6	Live protocol dashboard, auction terminal, and telemetry UI.
5	Empirical Simulation	Month 6–7	Monte Carlo experimental benchmark dataset and comparative plots.
6	Documentation & Defense	Month 7–8	Final B.Tech project synopsis, thesis dissertation, and defense slides.

CHAPTER 8

CONCLUSION

Decentralized lending protocols are foundational building blocks of the decentralized financial ecosystem, yet their reliance on first-come, first-served liquidations in public mempools has introduced severe systemic vulnerabilities. As demonstrated by recent empirical literature, Priority Gas Auctions cause 70% to 90% of gross liquidation profits to leak to consensus block builders, force distressed borrowers to forfeit excessive collateral via static penalty rates, and exacerbate secondary market price slippage through uncoordinated collateral dumping.

This project introduces **Reforge**, an end-to-end MEV-aware batch auction liquidation mechanism designed to replace predatory gas-bidding races with objective economic efficiency. By accumulating competing liquidator bids across a discrete, short auction window (1–2 blocks) and evaluating bids via the **Net Economic Value** metric:

$$NEV = V_{\text{gross}} - C_{\text{gas}} - C_{\text{slippage}} \quad (8.1)$$

Reforge ensures that liquidations are awarded to participants who execute with the lowest transaction overhead and minimal secondary market price impact.

The system is comprehensively architected across four modular tiers:

1. **EVM Smart Contracts:** Delivering robust collateral custody, lending pools, discrete auction lifecycles, and atomic settlement routines implemented in Solidity.
2. **High-Performance Backend Pipeline:** Providing real-time WebSocket block listening, asynchronous job queue scheduling via Redis/BullMQ, and off-chain bid ingestion and scoring.
3. **User and Liquidator Portal:** Offering a modern Next.js 16 web interface for protocol health telemetry and liquidator auction participation.

4. **Empirical Simulation Engine:** Enabling reproducible Monte Carlo benchmarking across historical crash regimes to quantify the precise trade-off between auction latency and protocol solvency.

By aligning the economic incentives of borrowers, protocols, and liquidators, Reforge demonstrates that application-level mechanism redesign can successfully mitigate MEV extraction. The project provides an open-source, academically grounded blueprint for building more resilient, fair, and capital-efficient lending architectures in decentralized finance.

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APPENDIX A

PLAGIARISM REPORT

A.1 Plagiarism Report

A.2 AI Content Report